

Mark Scheme

Practice Paper (Moderate)

Pearson Edexcel GCE AS Level

Economics A (8EC0)

Paper 01: Introduction to Markets and Market Failure

General marking guidance

- All candidates must receive the same treatment.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark scheme — not according to their perception of where the grade boundaries may lie.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification/indicative content will not be exhaustive.
- Crossed-out work should be marked unless the candidate has replaced it with an alternative response.

Section A

Question Number	Answer	Mark
1(a)	Knowledge 1, Application 1, Analysis 1 Explain the distinction between renewable and non-renewable resources. Knowledge/Analysis (2 marks) • Renewable resources can naturally replenish themselves over time / are not depleted by use (1) • Non-renewable resources are finite / limited in supply / cannot replenish in a usable timeframe (1) Application (1 mark) • Reference to solar power as renewable OR diesel/fossil fuel as non-renewable (1)	(3)

Question Number	Answer	Mark
1(b)	Knowledge 1 The only correct answer is C — A coffee roasting machine in a factory. A is incorrect — a barista is labour, not capital. B is incorrect — a coffee bean is land (a natural resource / raw material). D is incorrect — a consumer drinking coffee is consumption, not production.	(1)

Question Number	Answer	Mark
2(a)	Knowledge 1 1 mark for any acceptable definition of demand, e.g.: • The quantity of a good/service that consumers are willing AND able to buy at a given price over a given time period (1) • Willingness and ability of consumers to purchase a good at various prices (1)	(1)

Question Number	Answer	Mark
2(b)	Knowledge 1 The only correct answer is D — inward shift of the PPF. A is incorrect — point outside the original PPF, not feasible following loss of productive capacity. B is incorrect — shows a movement along the existing PPF (reallocation). C is incorrect — outward shift indicates growth, not destruction of productive capacity.	(1)

Question Number	Answer	Mark
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2(c)	Knowledge 1, Application 1 • New supply curve drawn and labelled showing a leftward/inward shift from S to S$\blacksquare$ (S2) (1) • New equilibrium identified with higher price (P$\blacksquare$/P2) and lower quantity (Q$\blacksquare$/Q2) clearly labelled (1)	(2)
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Question Number	Answer	Mark
3(a)	Knowledge 1 The only correct answer is C. A is incorrect — that describes a command economy. B is incorrect — that describes a pure free market economy. D is incorrect — even mixed economies have taxes and regulation.	(1)

Question Number	Answer	Mark
3(b)	Knowledge 1, Application 1, Analysis 1 Knowledge (1) • Vaccinations generate positive externalities of consumption / are a merit good which benefits society beyond the individual (1) • Asymmetric information — individuals underestimate the private benefits and may free-ride on others (1) Application (1) • Reference to vaccinations protecting 'society as a whole' / 'children and the elderly' from disease (1) Analysis (1) • Free market underprovides because external benefits are ignored / MSB > MPB / quantity is below social optimum (1) • Diagram showing positive consumption externality with welfare loss triangle (1 AN)	(3)

Question Number	Answer	Mark
4(a)	Knowledge 1 The only correct answer is C. Statement 1 is a factual, testable claim (positive). Statement 2 uses 'should' — a value judgement (normative).	(1)

Question Number	Answer	Mark
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4(b)	Knowledge 1, Application 1, Analysis 1 Knowledge (1) • Opportunity cost is the value of the next best alternative forgone when making an economic choice (1) Application (1) • Reference to the £2.5 billion of NHS spending / reduction in transport infrastructure spending (1) Analysis (1) • Because government resources are scarce, additional spending on the NHS requires a reduction elsewhere — transport infrastructure is the forgone alternative (1) • PPF diagram showing trade-off may be credited (1 AN max)	(3)
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Question Number	Answer	Mark
5(a)	Knowledge 1, Application 1 Knowledge (1) • $PED = \% \text{ change in quantity demanded} \div \% \text{ change in price}$ (1) Application (1) • $\% \text{ change in price} = (12.99 - 10.99) / 10.99 \times 100 = +18.2\%$ • $PED = -4\% \div 18.2\% \approx -0.22$ • Accept range -0.20 to -0.22 (or positive equivalent). Award 2 marks for correct value with working. Award 1 mark for correct formula or $\%$ change in price calculated correctly.	(2)

Question Number	Answer	Mark
5(b)	Knowledge 1 1 mark for any acceptable definition, e.g.: • PED measures the responsiveness of quantity demanded to a change in the price of the good (1)	(1)

Question Number	Answer	Mark
5(c)	Knowledge 1 The only correct answer is C — more substitutes makes demand more elastic. A is incorrect — fewer substitutes makes demand less elastic. B is incorrect — exclusive content reduces availability of close substitutes (less elastic). D is incorrect — long-term contracts lock consumers in, reducing elasticity.	(1)

Section B — Question 6

Question Number	Answer	Mark
6(a)	Knowledge 1, Application 2, Analysis 2 Knowledge (1) • 'Unintended substitution' is when consumers change to an alternative good that the policy did not target (1) Application (2) • Reference to consumers switching from sugary soft drinks to fruit juice / milkshakes (1) • Reference to the sugar tax (SDIL) being charged at 24p per litre on drinks with > 8g of sugar per 100ml / fruit juice not covered by the levy (1) Analysis (2) • Because the tax raises the relative price of sugary soft drinks, consumers substitute towards untaxed alternatives (1) • Yet these alternatives may contain just as much sugar — so the policy fails to reduce total sugar intake / market failure is not corrected (1) • Could be analysed as evidence of irrationality / asymmetric information about sugar content (1)	(5)

Question Number	Answer	Mark
6(b) KAA	Knowledge 2, Application 2, Analysis 2 (KAA 6) Indicative content — arguments that consumers may be acting irrationally: • Continued consumption despite known health damage (obesity, type 2 diabetes, tooth decay) — bounded self-control • Habit formation / addiction to sugar — consumers fail to weigh long-term costs against short-term satisfaction • Asymmetric information — children unaware of how much sugar drinks contain (extract reference) • Anchoring on familiar brand prices — consumers underweight the true cost increase • Evidence from Figure 1 — UK sales of sugar-sweetened drinks have fallen from 3,850 m litres (2016) to 2,610 m litres (2023), but many consumers still continue purchasing despite all the policy interventions Indicative content — arguments that consumers may be acting rationally: • Consumers maximise utility — full-sugar drinks provide more perceived enjoyment • Price rise of < 24p per litre is small relative to disposable income • Consumers may have low income elasticity / view sugary drinks as an affordable luxury • Habitual consumers gain time savings by buying familiar brands (rational shortcut)	(6)

Level	Mark	Descriptor
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Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

6(b) Evaluation (4)

Question Number	Answer	Mark
6(b) Eval	Evaluative points may include: • Consumers may simply lack information about the sugar content of drinks • Brand loyalty / habit makes price increases of 24p per litre too small to change behaviour • Low-income consumers face budget constraints that limit substitution • Continued purchase may be rational because the marginal utility from consumption still exceeds the marginal price • Reformulation has reduced sugar in many drinks — consumers may not perceive much health difference	(4)

Level	Mark	Descriptor
Level 1	1–2	Identification of generic evaluative comments without supporting evidence / reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
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6(c)	Knowledge 2, Application 2 Knowledge (2) • YED for low-sugar alternatives will be negative / they are inferior goods, since demand rises as income falls (1) • Coefficient is income-elastic ($YED > 1$) (1) Application (2) • % change in quantity = $(540 - 480) / 480 \times 100 = +12.5\%$ • % change in income = -2.1% • $YED \approx +12.5 / -2.1 \approx -5.95$ (1 + 1) Accept range -5.9 to -6.0. Award 2 application marks for a fully correct calculation; award 1 for the correct % change in quantity or correct formula applied. If the candidate gives only the coefficient with no explanation award maximum 2 marks.	(4)
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Question Number	Answer	Mark
6(d)	Knowledge 2, Application 2, Analysis 2 Knowledge (2) • Sugary drinks and low-sugar alternatives are substitutes — positive cross elasticity of demand (1) • Nudges shift consumer preferences without changing prices, working via choice architecture (1) Application (2) • Eye-level shelf placement increases purchases of low-sugar drinks by 3.1% (1) • Removing sugary drinks from checkout displays raises purchases of healthier options by 4.2% / traffic-light labels by 2.6% (1) Analysis (2) • As consumers switch to alternatives, demand for sugary drinks shifts inwards / leftwards ($D \rightarrow D_{\blacksquare}$) (1) • Warning labels reduce asymmetric information / make consumers more aware of the external costs of consumption (1) • Suitable demand diagram (D shifts inward for sugary drinks) may be credited (1)	(6)

Question Number	Answer	Mark
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6(e) KAA	Knowledge 3, Application 3, Analysis 3 (KAA 9) Private costs of consuming sugary drinks: • Price paid for the drink itself / dental treatment costs / out-of-pocket healthcare costs to the consumer • Forgone leisure / income from time spent ill External costs of consuming sugary drinks: • Cost to the NHS of treating obesity, type 2 diabetes and tooth decay (taxpayer burden) • Lost productivity / absenteeism in the wider economy • Welfare loss / DWL: $MSC > MPC$ and consumption beyond the socially optimum quantity ($Q_1 > Q^*$) • Diagram showing negative consumption externality with $MSB < MPB$ and welfare loss triangle credited • Children's exposure / future generation health costs NB: Level 3 responses must consider BOTH private and external costs of consumption of sugary drinks.	(9)
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Level	Mark	Descriptor
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Answer	Mark
6(e) Eval	Evaluation 6 • Difficulty of accurately valuing external costs (e.g. lifetime NHS cost of obesity) • Some external costs may be offset by tax revenues (£240 m raised by SDIL) that are recycled into school sports / breakfast clubs • Private benefits to consumers (utility, refreshment, energy) should be set against private costs • Costs may be progressive in the long run (reduced obesity), even if regressive in the short run for low-income consumers • Magnitude of external costs depends on PED — if inelastic, the social cost of continued consumption remains high • Substitution to other high-sugar products (juice, milkshakes) may mean external costs persist regardless of action against soft drinks	(6)

Level	Mark	Descriptor
Level 1	1–2	Identification of generic evaluative comments without supporting evidence / reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence / reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Answer	Mark
6(f) KAA	Knowledge 4, Application 4, Analysis 6 (KAA 14) Impacts of the SDIL on consumers and producers: • Price of sugary drinks rises by up to 24p per litre — consumer surplus falls and demand contracts (movement along D) • Producer surplus falls if PED is elastic; if PED is inelastic the burden falls more on consumers • Reformulation by major manufacturers (Coca-Cola, Britvic) leads to lower sugar content — partial market correction towards Q^* • £240 m of revenue raised in year 1 funds school sports and breakfast clubs (hypothecated revenue) • Incidence of the tax: consumer burden vs producer burden depends on relative elasticity of demand and supply — diagram required showing $S \rightarrow S + \text{tax}$, equilibrium moving from P, Q to $P_{\square}$, $Q_{\square}$ • Welfare gain triangle if the externality is internalised If no valid diagram is included, mark maximum Level 3 (10).	(14)

Level	Mark	Descriptor
Level 1	1–3	Displays isolated or imprecise knowledge. Generic or irrelevant information. Descriptive, no chains of reasoning.
Level 2	4–6	Some knowledge applied to context but lacks breadth. Two-stage chains of reasoning only.
Level 3	7–10	Accurate knowledge applied directly to the broad elements of the question. Coherent chains of reasoning but may lack balance.
Level 4	11–14	Precise knowledge applied with relevant examples fully integrated. Logical, balanced and coherent chains of reasoning.

Question Number	Answer	Mark
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6(f) Eval	Evaluation 6 • Effectiveness depends on PED — if demand for full-sugar drinks is inelastic, the tax has limited effect on quantity but raises revenue • Regressive impact on low-income households (tax falls disproportionately) • Unintended substitution to other sugary products (juice, milkshakes) limits public-health gains • Reformulation has meant only £240 m raised — far less than the £520 m originally projected • Government failure: difficult to set the rate at the exact level of the external cost • Alternative interventions (information provision, advertising bans) may be more cost-effective	(6)
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Level	Mark	Descriptor
Level 1	1–2	Identification of generic evaluative comments without supporting evidence / reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence / reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Answer	Mark
6(g) KAA	Knowledge 4, Application 4, Analysis 6 (KAA 14) Policy options to reduce market failure from sugary drinks consumption: • Indirect taxation (SDIL) — raises price, contracts demand, internalises external costs (diagram credited) • Subsidies for sugar-free alternatives — increases supply, lowers price, encourages substitution • Information provision — traffic-light labelling, public-health advertising — addresses asymmetric information • Regulation — limits on advertising to children, mandatory reformulation thresholds, restrictions on point-of-sale displays • Nudges (Extract C) — shelf placement, removing checkout displays, warning labels • Ban on sales to under-16s or in schools If no valid diagram is included, mark maximum Level 3 (10).	(14)

Level	Mark	Descriptor
Level 1	1–3	Isolated or imprecise knowledge; descriptive approach.
Level 2	4–6	Some knowledge applied to context but narrow.
Level 3	7–10	Accurate knowledge applied with chains of reasoning.
Level 4	11–14	Precise knowledge; logical and coherent chains of reasoning.

Question Number	Answer	Mark
6(g) Eval	Evaluation 6 • PED determines effectiveness of taxes / subsidies • Regressive impact of taxes on low-income consumers • Administrative costs of monitoring and enforcement • Information provision is limited if consumers exhibit bounded rationality • Opportunity cost of government spending on subsidies / advertising campaigns • Risk of government failure — distortion of price signals, unintended consequences (substitution, black markets) • Combinations of policies (Extract C) likely to be more effective than any single measure	(6)

Level	Mark	Descriptor
Level 1	1–2	Identification of generic evaluative comments without supporting evidence / reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence / reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.